SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I **SWETA R (192511408)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:Sweta R

Annexure A

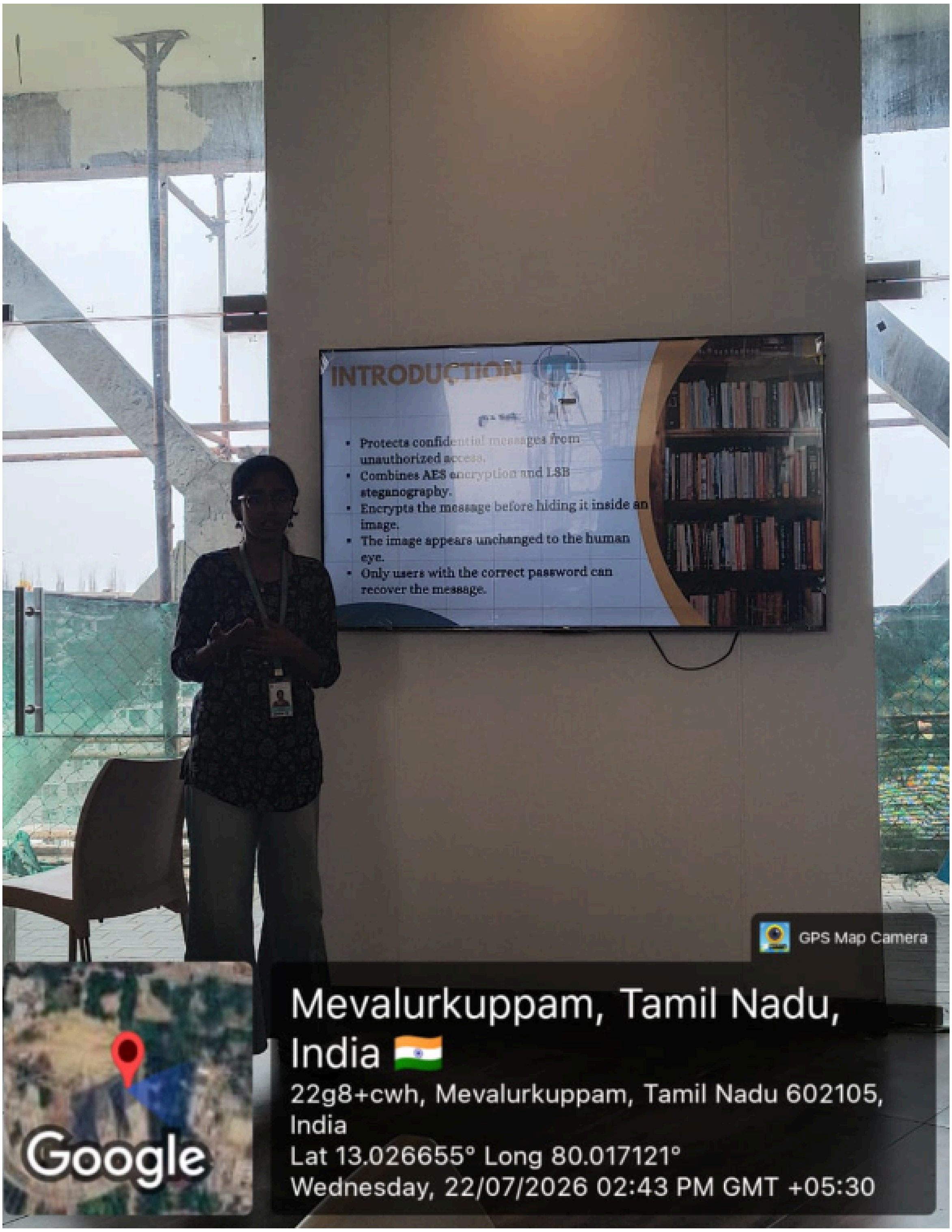
TECHNICAL PRESENTATION

“Steganography vs Cryptography: Hidden Security in the Digital World”

Steganography vs Cryptography: Hidden Security in the Digital World

Steganography and cryptography are two important techniques used to protect digital information. Cryptography secures data by converting it into an unreadable format using encryption, while steganography hides the existence of the data by embedding it within images, audio, or video files. Although both aim to ensure secure communication, they use different approaches. This topic compares their working principles, advantages, limitations, and real-world applications to understand how they contribute to digital security.

With the rapid growth of the internet, protecting sensitive information has become increasingly important. Cryptography ensures that only authorized users can access the original data, whereas steganography prevents attackers from even knowing that a secret message exists. These techniques are widely used in banking, military communication, cybersecurity, digital forensics, and secure online transactions. In many modern security systems, both methods are combined to provide an additional layer of protection. Understanding the differences between steganography and cryptography helps in selecting the appropriate technique for different security requirements.


INTRODUCTION
Protects confidential messages from unauthorized access.
Combines AES encryption and LSB steganography.
Encrypts the message before hiding it inside an image.
The image appears unchanged to the human eye.
Only users with the correct password can recover the message.
GPS Map Camera
Google
Mevalurkuppam, Tamil Nadu, India 🇮🇳
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026655° Long 80.017121°
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RUBRICS FOR CALCULATION:

Criteria	Weigh ta ge	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understa ndi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/techno lo gies	Good understa ndi ng with proper integration of most concepts	Basic understan din g; limited or partial integration	Poor understan din g; unable to integrate concepts
Experim ent al Design	30	Well-planned, systematic implementati on with optimal solution	Correct implemen tat ion with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonab le interpreta tio n	Basic analysis with limited insights	Poor or incorrect analysis of results
Documen tat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized document ati on with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n